Git & GitHub

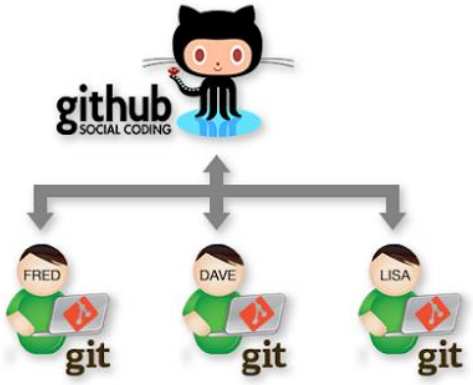
Git

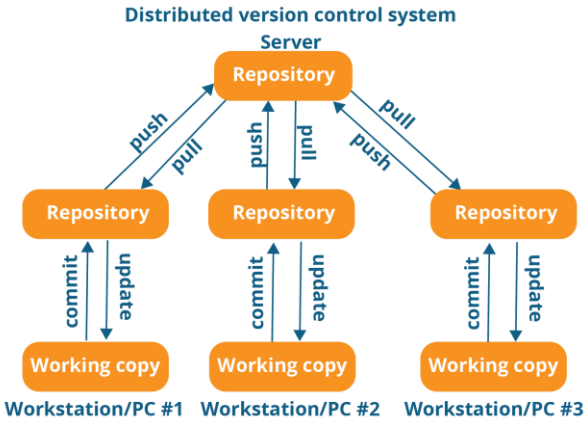
Git is a distributed version control system that helps developers track changes in code and collaborate on projects.

GitHub

GitHub is a cloud-based platform for hosting Git repositories, enabling collaboration and version control.

Git & GitHub Workflow







Steps for Git workflow:

1. Initialize a repository.
2. Configure user information.
3. Stage files for a commit.
4. Commit changes locally.
5. Link to a remote repository.
6. Push changes to the remote repository.
7. Pull latest changed from remote repository.
8. Delete local repository (Optional)

1. Create a New Local Git Repository

To initialize a new Git repository in a directory, navigate to the desired folder and execute the following command:

```
git init
```

2. Configure User Information (One-Time Setup)

Set your username and email for Git. This is typically a one-time setup for identifying the author of the commits. Replace "your name" and "your email" with your actual details:

```
git config --global user.name "your name"
git config --global user.email "your email"
```

3. Add Files or Folders to the Staging Area

To stage changes for the next commit, use the git add command. You can stage specific files, all files, or files matching certain patterns:

- **Add all files and folders to staging:**

  ```
  git add -A
  ```

- **Add a specific file:**

  ```
  git add filename
  ```

- **Add all files with a specific extension (e.g., Java files):**

  ```
  git add *.java
  ```

- **Add all files within a folder:**

```
git add foldername
git add .
```

4. Commit Changes to the Local Repository

After staging the files, commit them to your local repository with a descriptive commit message:

```
git commit -m "commit message"
```

5. Connect Local Repository to a Remote Repository (One-Time Setup)

Link your local repository to a remote repository using the git remote add command. Replace the example URL with the actual remote repository URL:

```
git remote add origin " https://github.com/pavanoltraining/myproject.git"
```

6. Push Changes to the Remote Repository

To upload your committed changes to the remote repository, use the git push command. Specify the remote name (origin) and the branch name (master for the main branch):

```
git push origin master
```

Need to pass token which is generated in GitHub.

7. Pull the latest Changes from the Remote Repository

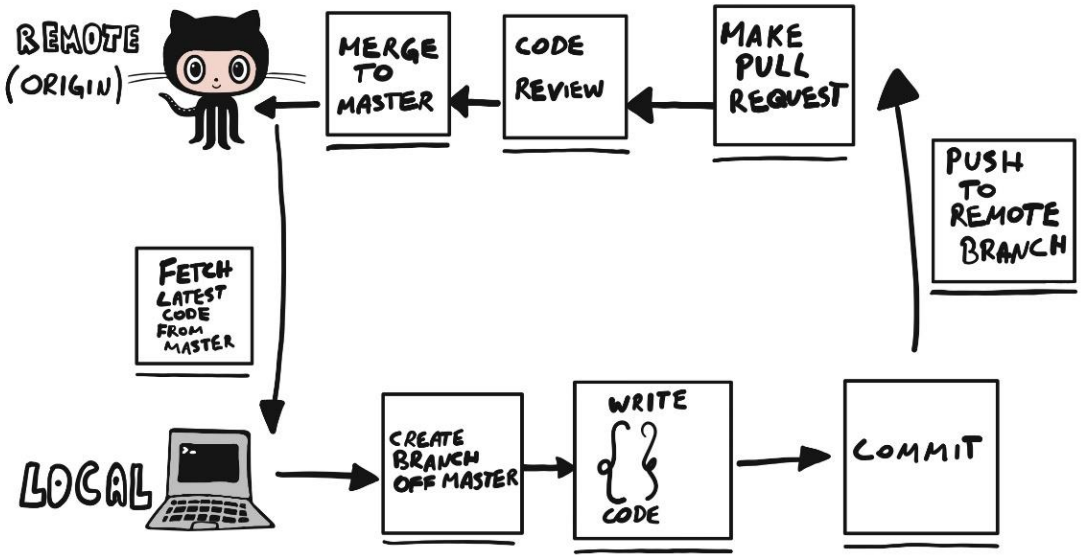
To get latest changes from the remote repository to local repository , you need to use git pull command.

```
git fetch origin master
(or)
git pull origin master
```

8. To Delete local repository

```
rm -rf
```

Branching & Merging



Step 1: Fetch the latest code from master (remote)

```
git fetch origin master
(or)
git pull origin master
(or)
git clone <<URL>>
```

Example:

```
git clone "https://github.com/pavanoltraining/myproject.git"
```

This updates your local master branch with the latest code from the remote repository.

Step 2: Create a new branch of master and switch to it.

```
git branch <branch name>          // creates a new branch
git checkout <branch name>        // switch to branch
git checkout -b <branch-name>      // single command
```

Example:

```
git branch Branch1

git checkout Branch1

git checkout -b Branch1   //single command for create a new branch and switch to it
```

This creates and switches to a new branch called **Branch1** based on the master branch.

Step 3: Write code and make changes

Make your changes to the files in your local repository. For example, you might edit file1.txt and create new files.

Step 4: Commit the changes

```
git add .

git commit -m "<commit-message>"
```

Example:

```
git add .

git commit -m "Added file2.txt & modified file1.txt"
```

This stages and commits all the changes in your branch.

Step 5: Push changes to the remote branch

```
git push origin <new-branch-name>
```

Example:

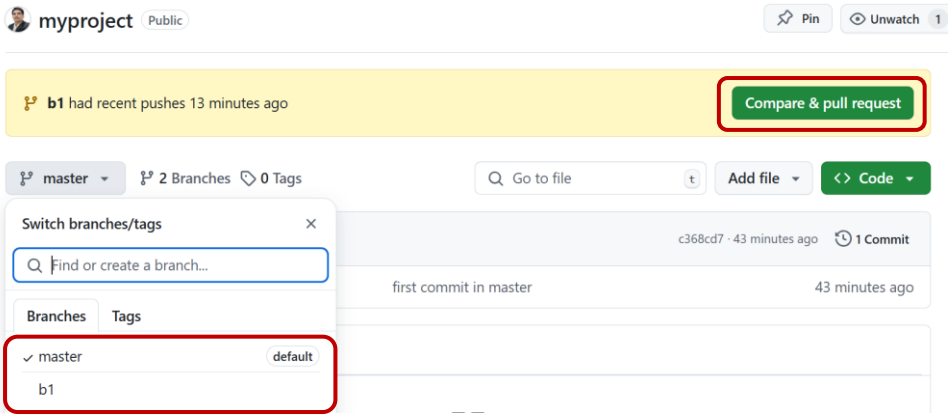
```
git push origin Branch1
```

This uploads your branch to the remote repository.

Step 6: Make a pull request

1. Go to your GitHub repository in a browser.
2. You'll see a message like *"Your branch has recent pushes"*. Click **Compare & Pull Request**.

3. Add a title and description for the pull request, then submit it.



Step 7: Code review

- Collaborators review the pull request, add comments, and suggest changes.
- If there are changes requested, make them in your local branch, commit them, and push them again:

```
git commit -m "Updated file1.txt based on review feedback"
git push origin Branch1
```

Step 8: Merge to master

After the code review is approved:

1. In GitHub, click **Merge Pull Request**.
2. Confirm the merge.
3. Your feature branch is now merged into master.

Optional Clean-Up: Delete the branch

After merging, you can delete the branch locally and remotely.

Delete local branch:

```
git branch -D <new-branch-name>
```

Delete remote branch:

```
git push origin --delete <new-branch-name>
```

Example:

```
git branch -D Branch1
git push origin --delete Branch1
```

Git Tutorials:

https://tinyurl.com/2ktjbscr